

AI Based Explainable Phishing Detection Assistant

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Abstract—Phishing detection systems increasingly rely on Artificial Intelligence (AI) to identify malicious emails, fraudulent websites, and QR-code-based phishing attacks while improving cybersecurity. With the rapid growth of cloud computing, digital banking, e-governance, and online communication, phishing attacks have become more sophisticated and difficult to detect using traditional rule-based methods. Although AI-based machine learning models achieve high detection accuracy, many operate as black-box systems that provide little explanation for their predictions, reducing user trust and limiting informed security decisions. Explainable Artificial Intelligence (XAI) addresses this challenge by generating human-understandable explanations while maintaining high detection performance. This paper presents an AI-Based Explainable Phishing Detection Assistant that integrates machine learning, LIME, SHAP, Large Language Models (LLMs), conversational AI, and QR-code (Quishing) detection into a unified framework for transparent phishing identification. The proposed framework analyses email content, sender information, URL characteristics, and QR codes to generate phishing risk scores, provide human-readable explanations, and recommend appropriate security actions. An interactive cybersecurity dashboard further improves user awareness by visualizing phishing trends and supporting informed decision-making. The proposed framework enhances transparency, user trust, and the reliability of AI-assisted phishing detection systems.

Index Terms— Artificial Intelligence (AI), Explainable Artificial Intelligence (XAI), Phishing Detection, Machine Learning, Large Language Models (LLMs), Conversational AI, QR-Code Phishing (Quishing), Email Security, Cybersecurity, Trustworthy AI

Phishing attacks have become one of the most serious cybersecurity threats affecting government organizations, educational institutions, financial services, businesses, and individual users. With the rapid adoption of cloud computing, digital banking, e-commerce, e-governance platforms, and online communication services, attackers continuously exploit deceptive emails, malicious websites, and QR codes to steal sensitive information such as login credentials, financial data, and personal information. These attacks have become increasingly sophisticated through the use of artificial intelligence and social engineering techniques, making them more difficult to detect using conventional security mechanisms.

Artificial Intelligence (AI)-based phishing detection significantly improves the identification of malicious emails and fraudulent websites by analysing email content, sender information, URL characteristics, and behavioural patterns. Machine learning and deep learning models can accurately distinguish phishing messages from legitimate communications, thereby reducing security risks and improving organizational cybersecurity.

However, most machine learning and deep learning models function as **black-box systems**, where users cannot understand how phishing predictions are generated. This lack of transparency reduces user confidence, particularly in security-critical environments where employees and individuals require clear evidence before trusting AI-generated warnings and making security decisions. **Explainable Artificial Intelligence (XAI)** has emerged as an effective solution by providing transparent reasoning behind AI predictions, thereby improving user trust, cybersecurity awareness, and informed decision-making. The integration of **Large Language Models (LLMs)** and conversational AI further enhances explainability by generating human-readable explanations, answering user queries, and recommending appropriate security actions. The proposed **AI-Based**

I. INTRODUCTION

Explainable Phishing Detection Assistant combines explainable machine learning, conversational AI, QR-code (Quishing) detection, and cybersecurity education into a unified framework for developing transparent, reliable, and trustworthy phishing detection systems.

II. EXPLAINABLE AI FOR PHISHING DETECTION

A. Importance of Explainability

Traditional Artificial Intelligence (AI) models primarily focus on achieving high phishing detection accuracy without explaining the reasoning behind their predictions. In phishing detection, this limitation reduces user confidence because individuals and cybersecurity analysts cannot verify whether an email has been classified as phishing based on genuine malicious indicators or irrelevant patterns. Explainable Artificial Intelligence (XAI) improves transparency by identifying the important email characteristics, URL attributes, sender information, and linguistic features that contribute to a phishing prediction. This enables users to understand, validate, and confidently act on AI-generated security recommendations, thereby improving cybersecurity awareness and trust in intelligent phishing detection systems.

Table I. Benefits of Explainable AI in Phishing Detection

Feature	Benefit
Transparent Predictions	Explains why an email classified as phishing
Feature Importance	Identifies the most influential phishing indicators
User Trust	Increases confidence in AI-generated alerts
Cybersecurity Awareness	Educates users about phishing characteristics
Decision Support	Assists users in taking appropriate security actions

B. Email Feature Analysis

Phishing detection systems continuously analyse email features such as sender domain reputation, email headers, URL characteristics, domain age, embedded hyperlinks, QR codes, attachment information, language patterns, urgency keywords, and brand impersonation indicators. Before phishing classification, these data are cleaned, normalized, and processed to remove redundant information and improve feature quality. Feature extraction techniques identify meaningful characteristics from email content and URLs, enabling AI models to

distinguish phishing emails from legitimate communications accurately. Explainability methods further determine which extracted features have the greatest influence on the final phishing prediction, making the detection process transparent, understandable, and reliable for both cybersecurity professionals and general users.

Table II. Common Features Used for Phishing Detection

Feature Category	Description
Sender Information	Domain reputation, SPF/DKIM authentication, sender authenticity
URL Features	URL length, HTTPS usage, redirects, suspicious domains
Email Content	Urgent language, spelling errors, credential requests
Attachment Analysis	Malicious attachments and executable files
QR-Code Features	Embedded URLs and QR-code legitimacy
Behavioural Features	Brand impersonation and social engineering patterns

III. EXPLAINABILITY TECHNIQUES FOR PHISHING DETECTION

A. Local Interpretable Model-Agnostic Explanations (LIME)

Local Interpretable Model-Agnostic Explanations (LIME) is a widely used Explainable Artificial Intelligence (XAI) technique that explains the predictions of machine learning models regardless of their architecture. Instead of explaining the entire model, LIME focuses on a single prediction by approximating the model with a simple interpretable model around the selected email instance. This helps users understand why an email has been classified as phishing.

In phishing detection, LIME identifies important features such as suspicious URLs, sender reputation, urgency keywords, credential requests, and QR codes that influence the prediction. These explanations improve transparency, increase user trust, and support informed cybersecurity decisions. However, LIME explains only individual predictions, and its results may vary depending on the sampling process.

B. SHapley Additive exPlanations (SHAP)

SHapley Additive exPlanations (SHAP) is an Explainable AI technique based on cooperative game theory that measures the contribution of each input feature to the final prediction. Unlike LIME, SHAP provides both local and global explanations, making it suitable for understanding individual predictions as well as overall model behaviour.

For phishing detection, SHAP highlights the importance of features such as sender reputation, URL characteristics, domain age, email headers, attachments, and suspicious language patterns. These feature contribution scores improve model transparency, support cybersecurity analysis, and enhance user confidence in AI-based phishing detection.

C. Large Language Models (LLMs) for Explainable Phishing Detection

Large Language Models (LLMs) improve Explainable AI by converting machine learning predictions into simple, human-readable explanations. Instead of displaying only technical feature importance values, LLMs explain why an email is considered suspicious and recommend appropriate security actions.

In the proposed phishing detection assistant, the LLM interprets the phishing prediction and generates conversational explanations describing suspicious URLs, sender behaviour, credential requests, QR-code threats, and social engineering indicators. This interactive approach improves cybersecurity awareness, user trust, and decision-making for both technical and non-technical users.

IV. AI-BASED EXPLAINABLE PHISHING DETECTION FRAMEWORK

The proposed **AI-Based Explainable Phishing Detection Framework** consists of four major stages: **Email Data Acquisition and Preprocessing**, **AI-Based Phishing Classification**, **Explainability Module**, and **User Guidance and Decision Support**. The overall framework is designed to detect phishing attacks accurately while providing transparent explanations that help users understand the reasons behind each prediction.

A. Email Data Acquisition and Preprocessing

The first stage of the proposed framework is **Email Data Acquisition and Preprocessing**. Emails are collected from mail servers, user inboxes, or publicly available phishing datasets. Each email contains valuable information such as sender details, email headers, URLs, embedded hyperlinks, QR codes, attachments, and message content.

Before classification, the collected email data undergo preprocessing to improve data quality. During this stage, duplicate emails are removed, missing values are handled, URLs and QR codes are extracted, text is normalized, and irrelevant information is filtered. Feature extraction is then performed to identify phishing indicators such as suspicious URLs, sender reputation, domain age, urgency keywords, credential requests, spelling errors, and brand impersonation. The extracted features form the input for the phishing detection model.

B. AI-Based Phishing Classification

In the second stage, the extracted features are analysed using machine learning algorithms to determine whether an email is legitimate or phishing. The classification model learns phishing patterns from historical email datasets and evaluates multiple indicators, including sender authenticity, URL characteristics, email content, attachments, and behavioural patterns.

Based on these features, the model generates a phishing risk score and classifies the email as **Legitimate** or **Phishing**. This automated analysis enables early detection of phishing attacks and reduces the possibility of users interacting with malicious emails.

C. Explainability Module

After the phishing classification is completed, the **Explainability Module** interprets the prediction using Explainable Artificial Intelligence techniques such as **LIME**, **SHAP**, and **Large Language Models (LLMs)**. Instead of displaying only the classification result, the module identifies the most influential phishing indicators that contributed to the prediction.

The generated explanation highlights factors such as suspicious URLs, domain reputation, urgency language, credential requests, embedded QR codes, and sender behaviour. The LLM further converts these technical findings into clear, human-readable explanations, enabling both technical and non-technical users to understand why an email has been classified as phishing.

D. User Guidance and Security Decision Support

The final stage presents the phishing prediction and explanation through an interactive dashboard and conversational assistant. Users receive the phishing risk score, detailed explanations, and practical security recommendations such as avoiding suspicious links, deleting malicious emails, reporting phishing attempts, or verifying the sender before responding.

The dashboard also visualizes phishing statistics, common attack indicators, and phishing trends to improve cybersecurity awareness. By combining accurate phishing detection with transparent explanations and user guidance, the proposed framework enhances trust in AI-based security systems and supports informed cybersecurity decision-making.

V. APPLICATIONS OF AI-BASED EXPLAINABLE PHISHING DETECTION

The proposed **AI-Based Explainable Phishing Detection Assistant** has wide applications across government organizations, educational institutions, financial services, healthcare systems, businesses, and e-commerce platforms. These sectors rely heavily on email communication and online services, making them frequent targets of phishing attacks. The proposed framework analyses email content, sender information, URL characteristics, attachments, and QR codes to accurately identify phishing attempts while providing human-readable explanations for each prediction.

By integrating Explainable Artificial Intelligence (XAI), Large Language Models (LLMs), and conversational AI, the system enables users to understand why an email is classified as phishing and recommends appropriate security actions. The framework also supports cybersecurity awareness by visualizing phishing trends, common attack patterns, and user risk levels through an interactive dashboard. As a result, organizations can reduce successful phishing attacks, improve user trust, strengthen security awareness, and support informed cybersecurity decision-making.

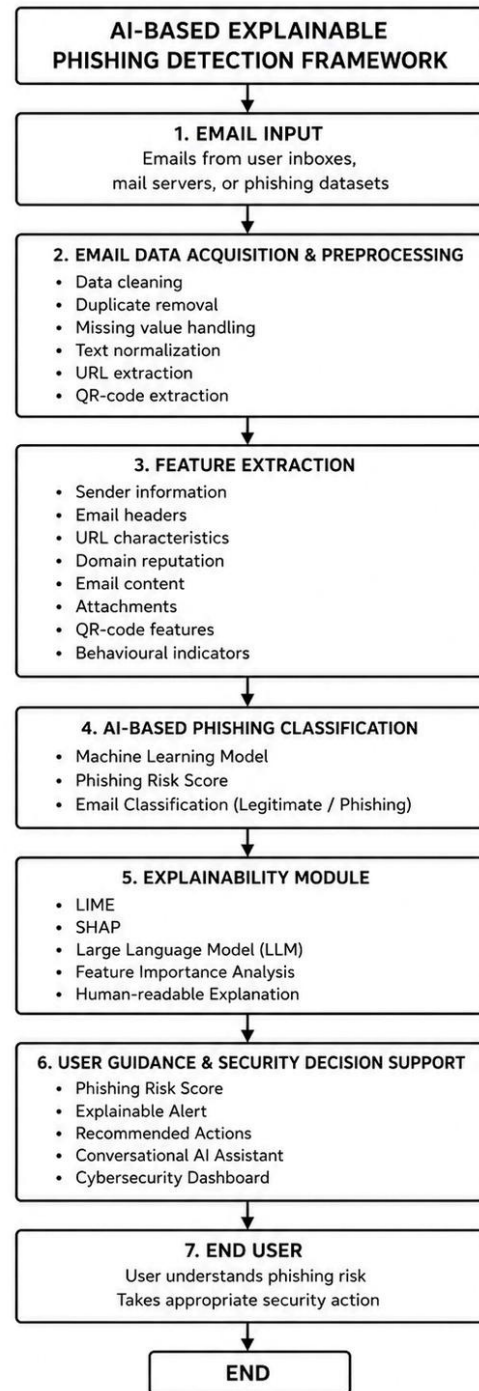


Fig. 1. Workflow of the Proposed AI-Based Explainable Phishing Detection Framework.

VI. TOOLS AND TECHNOLOGIES USED

Table I. Technologies Used in the Proposed AI-Based Explainable Phishing Detection Framework

Technology	Purpose
Email Datasets	Train and evaluate phishing detection models
Python	Data preprocessing and model implementation
Machine Learning Models	Classify emails as phishing or legitimate
LIME	Generate local explanations for phishing predictions
SHAP	Analyse feature contributions to model predictions
Large Language Models (LLMs)	Produce human-readable explanations and conversational responses
QR-Code Detection Libraries	Detect and analyse malicious QR codes (Quishing)
Streamlit/Web Dashboard	Display phishing analysis and cybersecurity awareness

The proposed framework integrates several technologies to provide accurate and explainable phishing detection. Email datasets are used to train and evaluate machine learning models, while Python supports data preprocessing, feature extraction, and model development. Machine learning algorithms classify suspicious emails based on multiple phishing indicators. LIME and SHAP improve model transparency by explaining the influence of important features on phishing predictions. Large Language Models generate natural-language explanations and support conversational interaction with users. QR-code detection libraries analyse embedded QR codes for hidden phishing URLs, and a web-based dashboard presents phishing alerts, explanations, and cybersecurity awareness statistics in an intuitive manner.

VII. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

Although Explainable Artificial Intelligence significantly improves transparency in phishing detection, several challenges remain. Modern phishing attacks continuously evolve using AI-generated content, domain spoofing, QR-code phishing, and sophisticated social engineering techniques, making them increasingly difficult to detect. Balancing high detection accuracy with meaningful explanations also remains a challenge, particularly for deep learning models. In addition, explanation quality may vary depending on the underlying machine learning

model and the prompting strategy used for Large Language Models.

Future research should focus on developing hybrid AI models that combine high detection performance with consistent and reliable explanations. Integrating real-time threat intelligence, multilingual phishing detection, adaptive learning, and personalized security recommendations can further enhance the effectiveness of phishing detection systems. Future work may also extend the framework to detect phishing attacks across SMS, instant messaging applications, and social media platforms while preserving user privacy and improving explainability.

VIII. CONCLUSION

The proposed **AI-Based Explainable Phishing Detection Assistant** combines machine learning, Explainable Artificial Intelligence, Large Language Models, conversational AI, and QR-code detection into a unified framework for transparent phishing detection. Unlike conventional black-box models, the proposed system provides clear explanations that enable users to understand phishing predictions, verify suspicious email characteristics, and make informed security decisions. The integration of conversational guidance and cybersecurity awareness further improves user trust and supports effective protection against phishing attacks.

As digital communication continues to expand, explainable and trustworthy AI will play an increasingly important role in protecting individuals and organizations from evolving cyber threats. Future developments should focus on real-world deployment, continuous model improvement, multilingual support, and integration with enterprise email security systems to build reliable, transparent, and intelligent phishing detection solutions

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